

JEE MAIN REVISION MASTER: ROTATIONAL MOTION 2026 2 QUESTIONS 3.33% WEIGHTAGE

↑99.4%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Rotational Motion - Core Application	13 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Rotational Motion - Core Application (High Yield — 13 PYQs)

Core Principle: Key problem pattern in Kcet Rotational Motion.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The angular momentum of a moving body remains constant, if..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1=4\text{kgm}^2, r_1=20\text{cm}$, $I_2=20\text{kgm}^2$ and $r_2=30\text{cm}$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?

- (A) 6 h
- (B) 18 h
- (C) 24 h
- (D) 30 h

Correct Answer: None

Solution Analysis: When the earth contracts suddenly, no external torque is acting on it. So its angular momentum remains conserved. Angular momentum is $L = I\omega$. For a spherical body, $I \propto R^2$. If the radius becomes half, $R' = (R / 2)$. So the new moment of inertia becomes $I' = I((1 / 2))^2 = (I / 4)$. Using conservation of angular momentum, $I\omega = I'\omega'$. $I\omega = (I / 4)\omega'$. $\omega' = 4\omega$. So the earth will rotate 4 times faster. Time period is inversely proportional to angular velocity, so $T' = (T / 4)$. Present duration of a day is $T = 24\text{h}$. Hence, $T' = 6\text{h}$.